Software Engineering

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Name : Uthkarsha

SRN : PES1UG24AM413

Section : AIML G

Project : Micro-Lending & Peer Credit Risk Assessor

SCRUM-5 – Loan Application & Risk Assessment

Description:

Enables Borrowers to submit loan applications with complete financial disclosures, and allows the Risk Scoring Engine to compute a credit risk tier (Low/Medium/High) using debt-to-income ratio, repayment history, and employment stability. Incomplete applications are automatically rejected.

User Stories:

- SCRUM-10 – Submit Loan Application — 5 points
- SCRUM-12 – Compute Credit Risk Tier — 8 points
- SCRUM-11 – Reject Incomplete Applications — 3 points

Total Story Points: 16

SCRUM-6 – Peer Lending Marketplace

Description:

Enables Peer Lenders to browse open loan requests, track funding status in real time, and commit partial or full funding toward a loan request.

User Stories:

- SCRUM-13 – Browse Loan Requests — 5 points
- SCRUM-15 – Track Funding Status — 3 points
- SCRUM-14 – Fund a Loan Request — 8 points

Total Story Points: 16

SCRUM-7 – Disbursement & Repayment Scheduling

Description:

Once a loan is fully funded, auto-generates a structured repayment schedule (amount, due dates, principal/interest breakdown) and disburses funds to the Borrower.

User Stories:

- SCRUM-16 – Generate Repayment Schedule — 8 points
- SCRUM-17 – Disburse Funds — 5 points

Total Story Points: 13

SCRUM-8 – Repayment Monitoring & Default Management

Description:

Continuously monitors repayment due dates, automatically sends default warning notifications to Borrowers and Peer Lenders when payments are missed, and allows Borrowers to request loan restructuring.

User Stories:

- SCRUM-18 – Monitor Repayment Due Dates — 5 points
- SCRUM-19 – Send Default Warning — 5 points
- SCRUM-20 – Request Loan Restructuring — 3 points

Total Story Points: 13

SCRUM-9 – Security & Transaction Integrity

Description:

Ensures all repayment and fund-transfer transactions execute safely as ACID-compliant database transactions, and that borrower financial and personal (PII) data is encrypted at rest and in transit.

User Stories:

- SCRUM-21 – ACID-Safe Repayment Transactions — 8 points
- SCRUM-22 – Encrypt Financial & PII Data — 5 points

Total Story Points: 13

Sprint Execution

A one-week Sprint 1 was created in Jira covering SCRUM-5 (Loan Application & Risk Assessment) and SCRUM-9 (Security & Transaction Integrity), since risk scoring and transaction security are the foundational capabilities the rest of the platform depends on. The sprint was started and the selected stories were progressed through the Agile workflow from To Do → In Progress → Done. All planned Sprint 1 stories were completed. A second one-week sprint was then used to complete the remaining marketplace, disbursement/repayment scheduling, and default management functionality (SCRUM-6, SCRUM-7 and SCRUM-8).

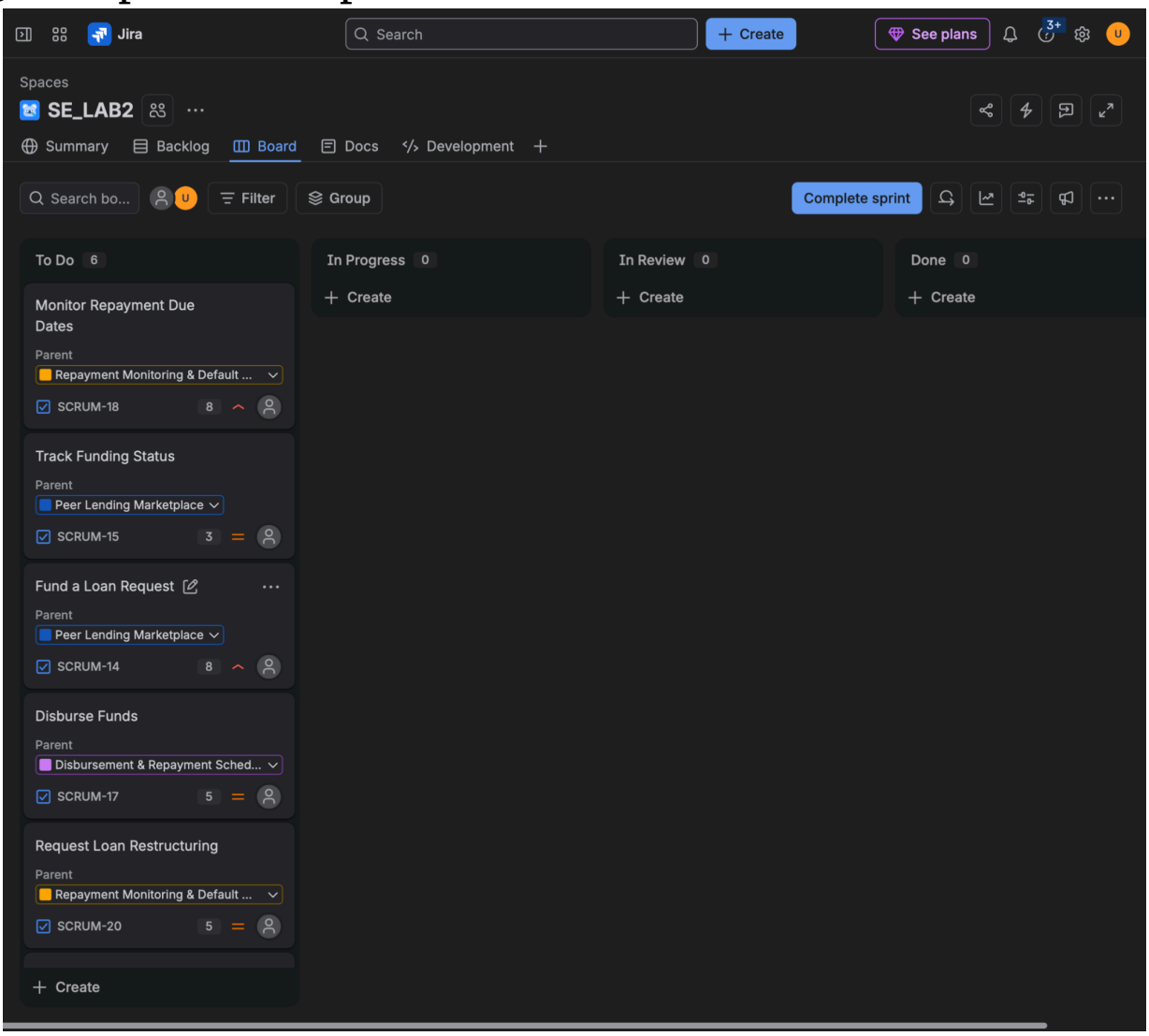
Story Point Estimation

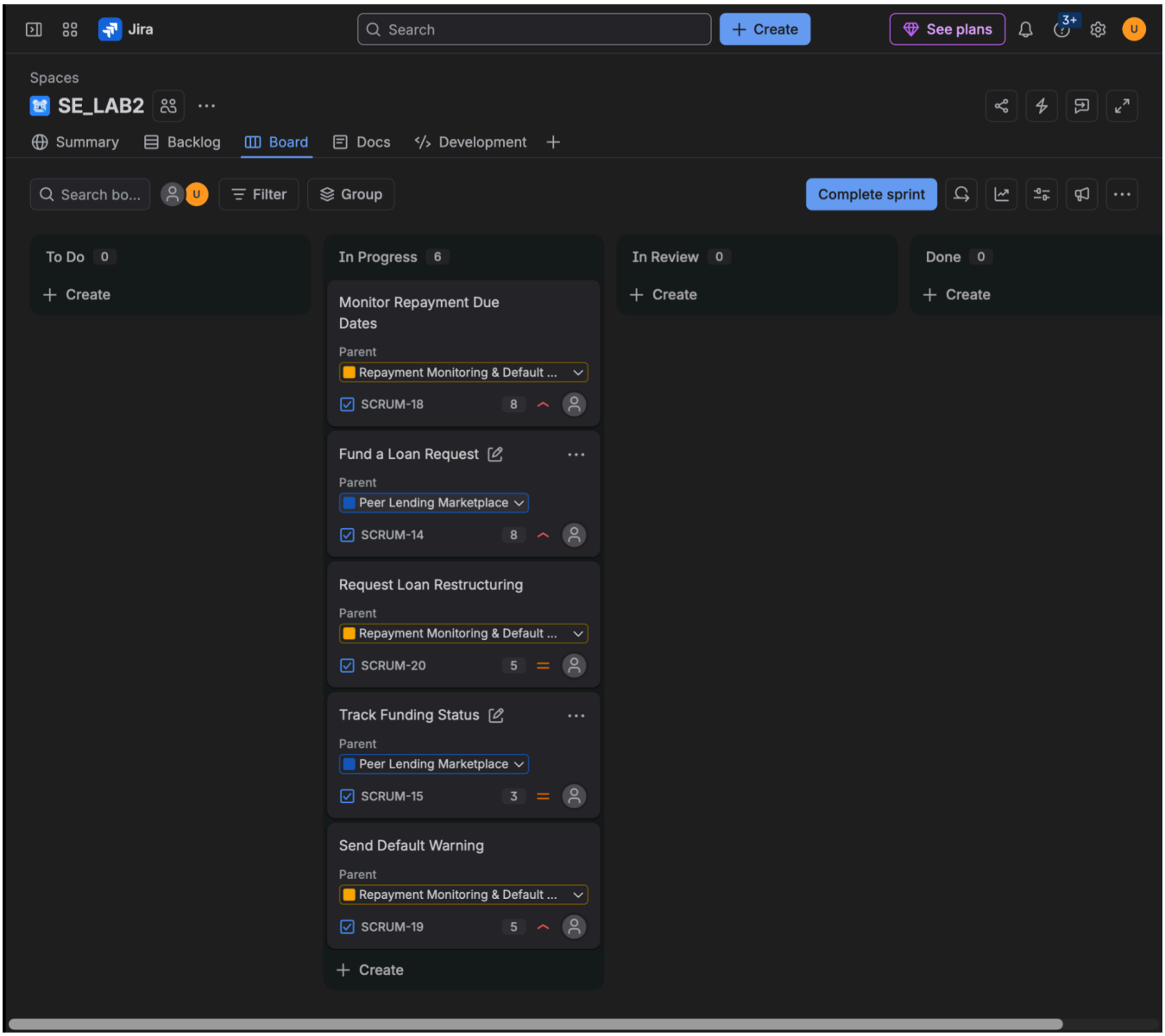
The lab requires assigning story points using the Fibonacci sequence. The story points were estimated based on the relative effort, complexity and uncertainty associated with each user story.

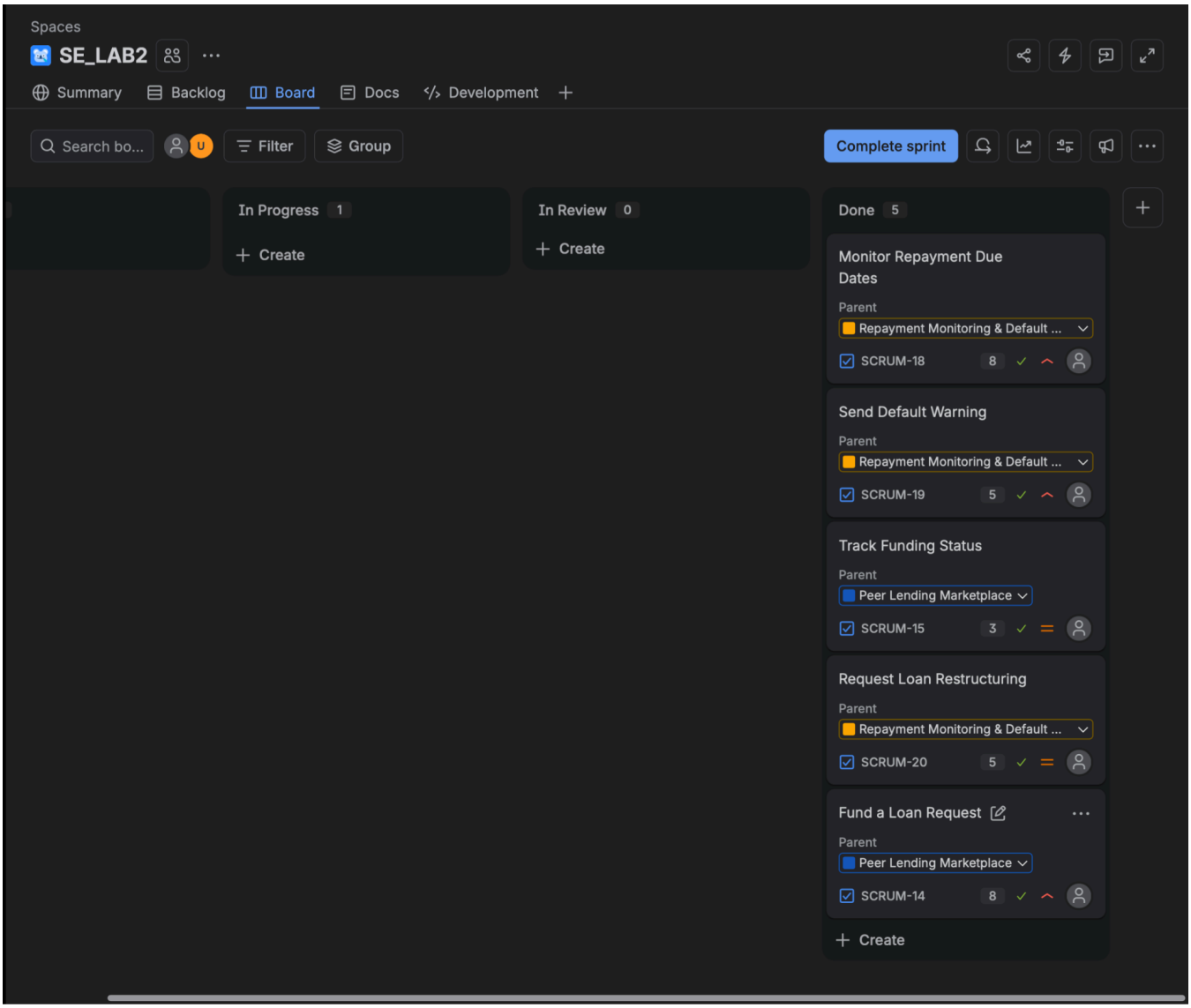
Story Points:

Story	Story Points
SCRUM-10 – Submit Loan Application	5
SCRUM-12 – Compute Credit Risk Tier	8
SCRUM-11 – Reject Incomplete Applications	3
SCRUM-13 – Browse Loan Requests	5
SCRUM-15 – Track Funding Status	3
SCRUM-14 – Fund a Loan Request	8
SCRUM-16 – Generate Repayment Schedule	8
SCRUM-17 – Disburse Funds	5
SCRUM-18 – Monitor Repayment Due Dates	5
SCRUM-19 – Send Default Warning	5
SCRUM-20 – Request Loan Restructuring	3
SCRUM-21 – ACID-Safe Repayment Transactions	8
SCRUM-22 – Encrypt Financial & PII Data	5
Total	**71**

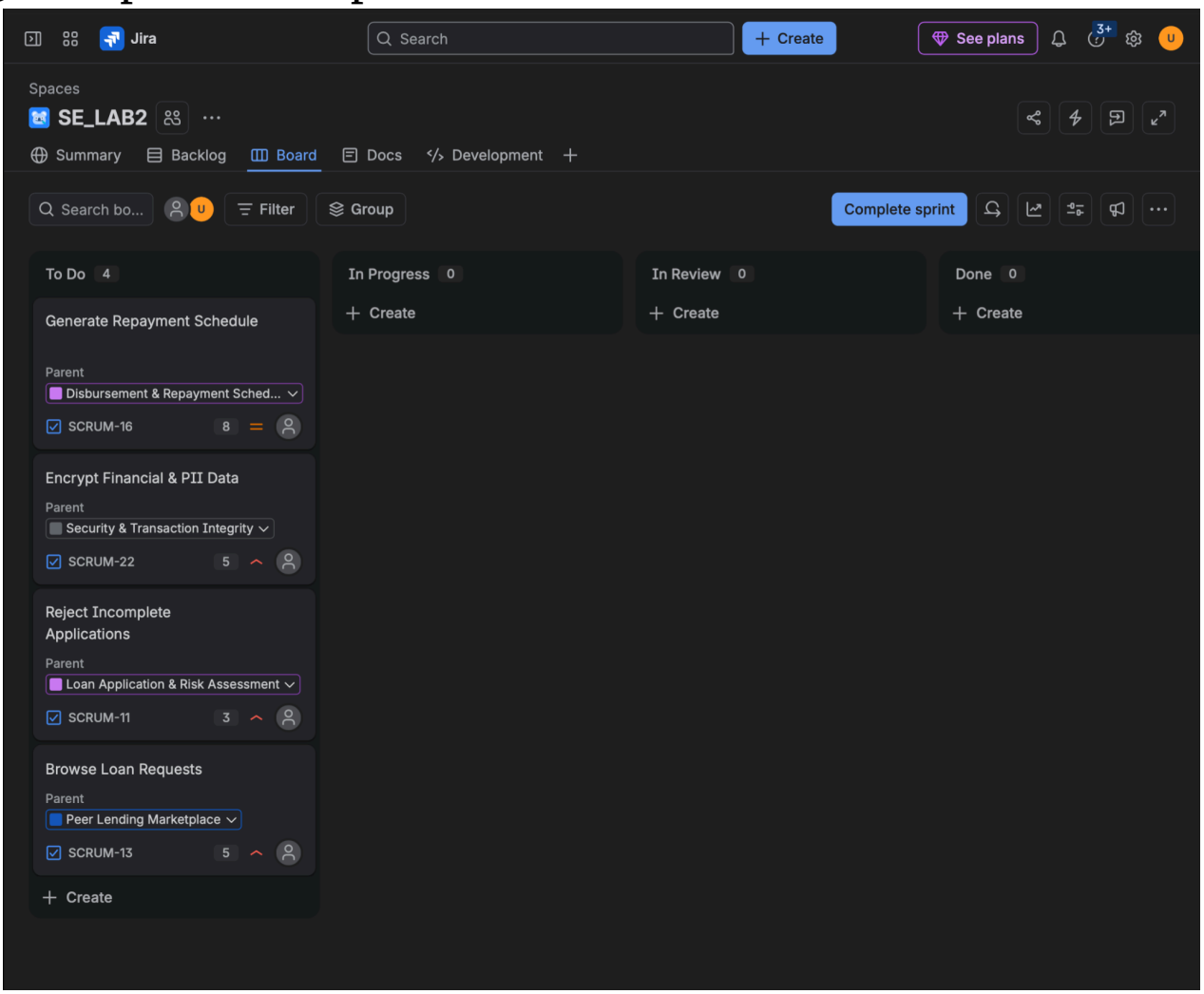
JIRA Sprint 1 Completion Screenshot

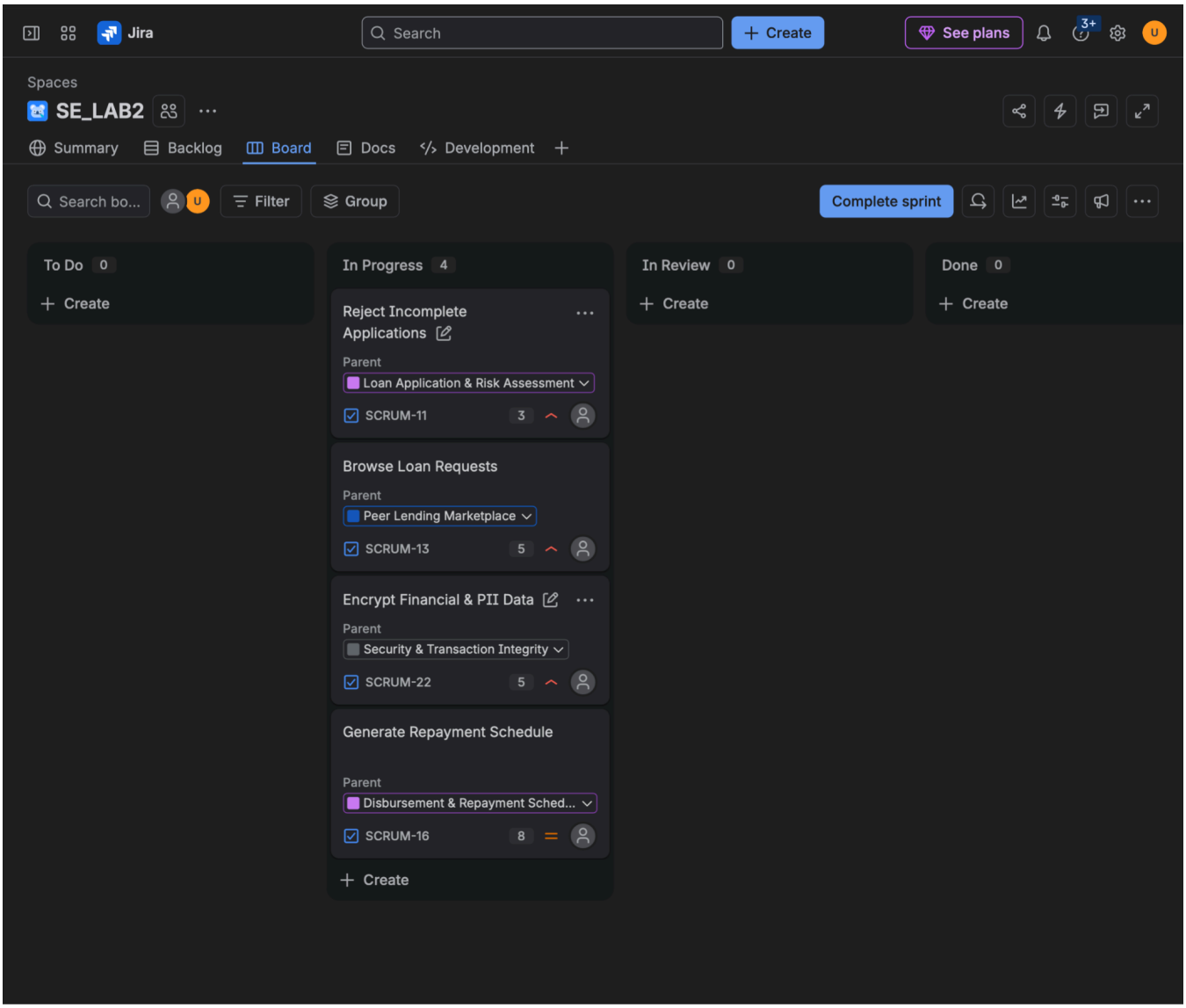


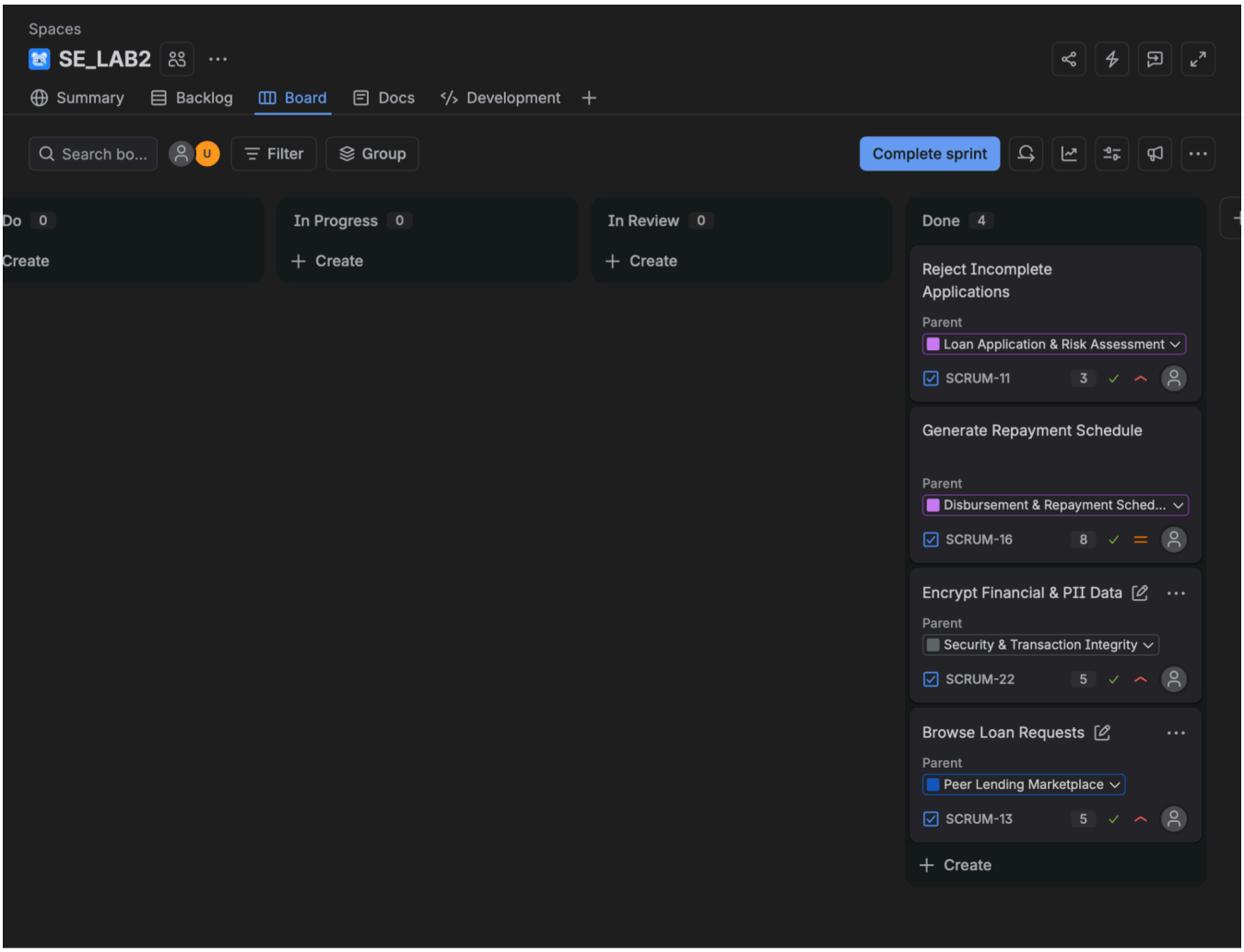

Jira
Search
Create
See plans
Spaces
SE_LAB2
Summary
Backlog
Board
Docs
Development
Search bo...
Filter
Group
Complete sprint
To Do 0
Create
In Progress 6
Monitor Repayment Due Dates
Parent
Repayment Monitoring & Default ...
SCRUM-18 8
Fund a Loan Request
Parent
Peer Lending Marketplace
SCRUM-14 8
Request Loan Restructuring
Parent
Repayment Monitoring & Default ...
SCRUM-20 5
Track Funding Status
Parent
Peer Lending Marketplace
SCRUM-15 3
Send Default Warning
Parent
Repayment Monitoring & Default ...
SCRUM-19 5
Create
In Review 0
Create
Done 0
Create


Spaces
SE_LAB2
Summary
Backlog
Board
Docs
Development
Search bo...
Filter
Group
Complete sprint
In Progress 1
Create
In Review 0
Create
Done 5
Monitor Repayment Due Dates
Parent
Repayment Monitoring & Default ...
SCRUM-18 8
Send Default Warning
Parent
Repayment Monitoring & Default ...
SCRUM-19 5
Track Funding Status
Parent
Peer Lending Marketplace
SCRUM-15 3
Request Loan Restructuring
Parent
Repayment Monitoring & Default ...
SCRUM-20 5
Fund a Loan Request
Parent
Peer Lending Marketplace
SCRUM-14 8
Create

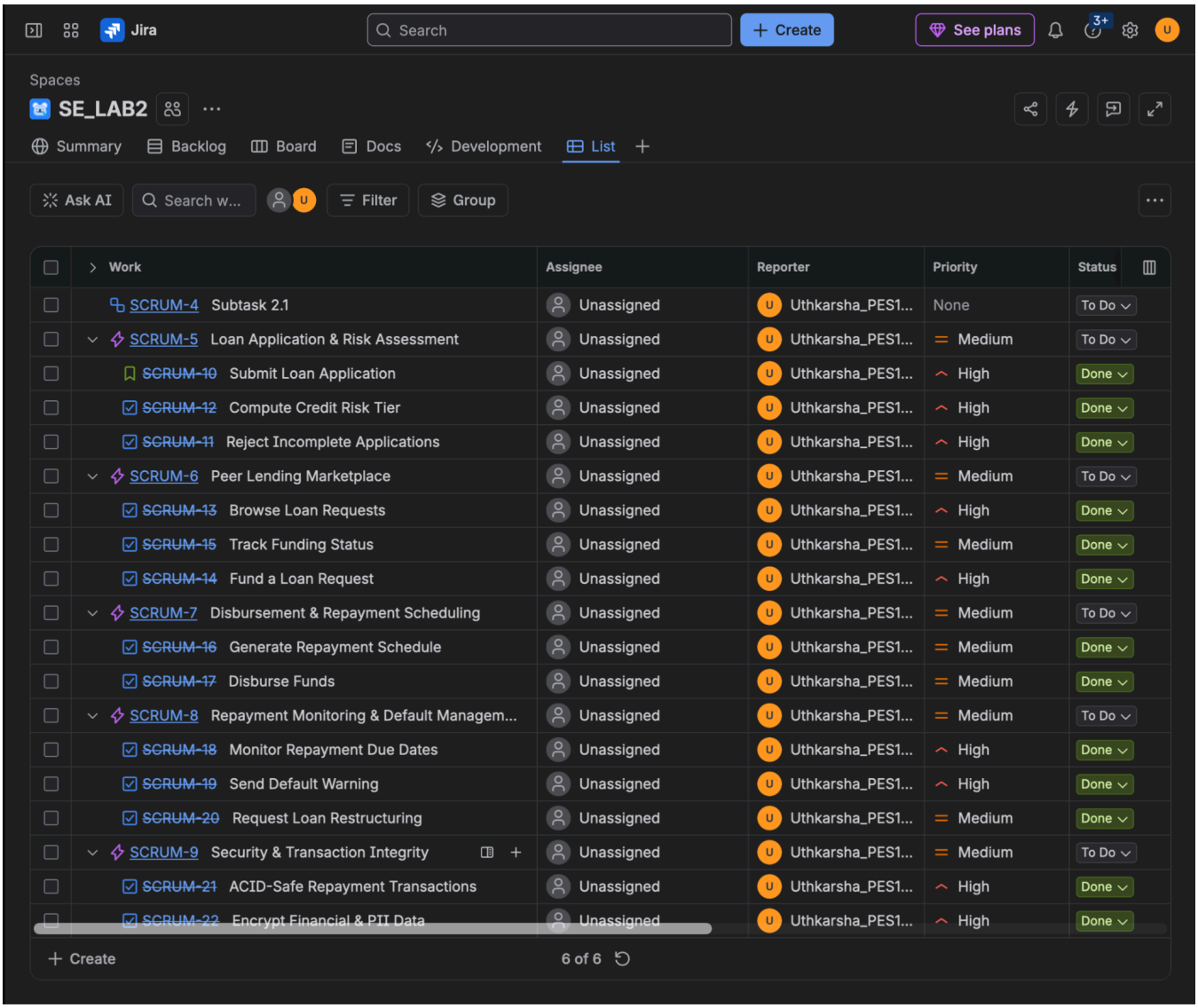
JIRA Sprint 2 Completion Screenshot




Jira
Search
Create
See plans
Spaces
SE_LAB2
Summary
Backlog
Board
Docs
Development
Search bo...
Filter
Group
Complete sprint
To Do 0
Create
In Progress 4
Reject Incomplete Applications
Parent
Loan Application & Risk Assessment
SCRUM-11
3
Browse Loan Requests
Parent
Peer Lending Marketplace
SCRUM-13
5
Encrypt Financial & PII Data
Parent
Security & Transaction Integrity
SCRUM-22
5
Generate Repayment Schedule
Parent
Disbursement & Repayment Sched...
SCRUM-16
8
Create
In Review 0
Create
Done 0
Create


Spaces
SE_LAB2
Summary
Backlog
Board
Docs
Development
Search bo...
Filter
Group
Complete sprint
Do 0
Create
In Progress 0
Create
In Review 0
Create
Done 4
Reject Incomplete Applications
Parent
Loan Application & Risk Assessment
SCRUM-11
3
Generate Repayment Schedule
Parent
Disbursement & Repayment Sched...
SCRUM-16
8
Encrypt Financial & PII Data
Parent
Security & Transaction Integrity
SCRUM-22
5
Browse Loan Requests
Parent
Peer Lending Marketplace
SCRUM-13
5
Create

JIRA All Epics and Stories with Story Points

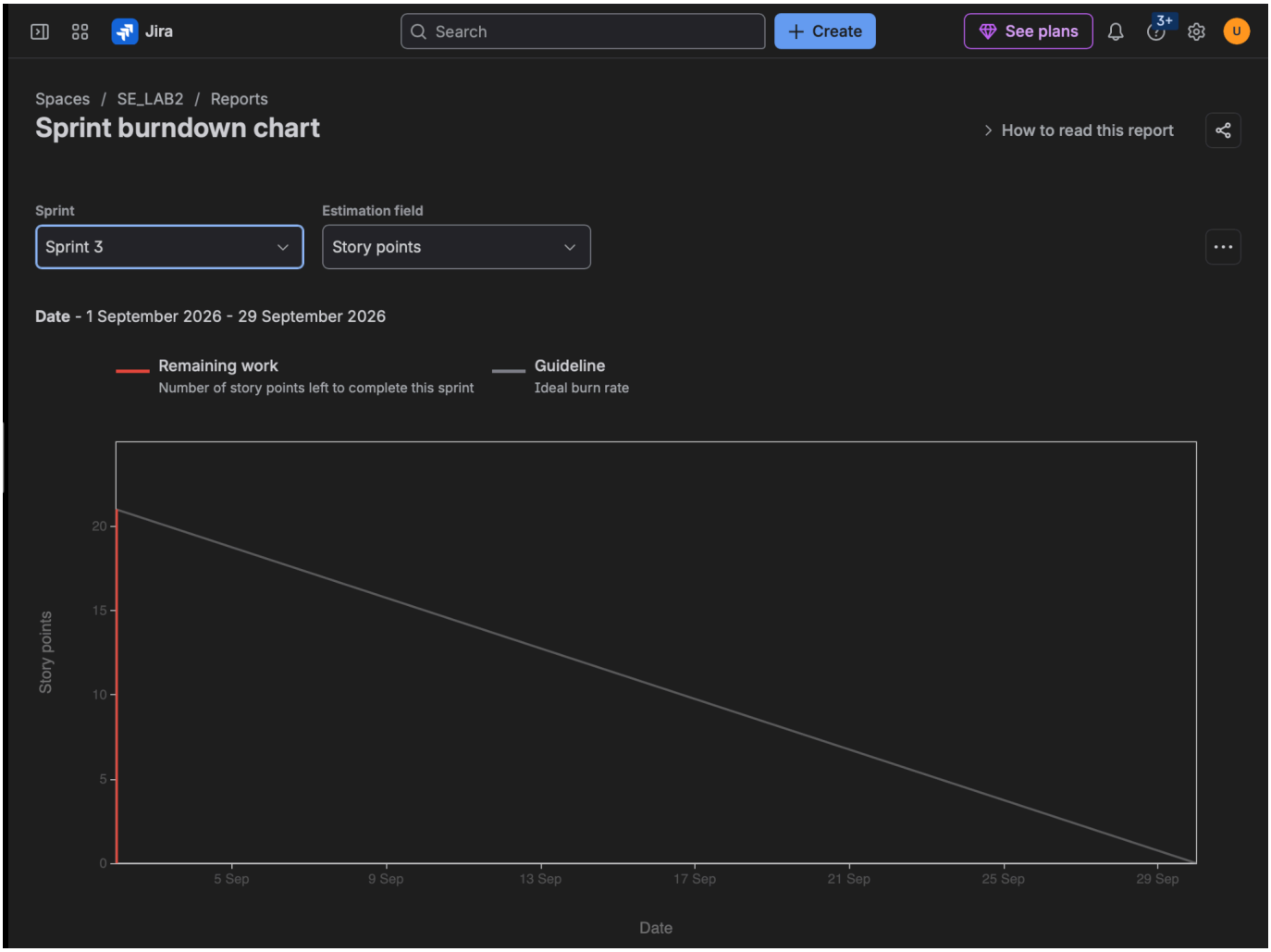


Burndown Chart

Sprint 1:



Sprint 2:


Jira
Search
Create
See plans
Spaces / SE_LAB2 / Reports
Sprint burndown chart
How to read this report
Sprint
Sprint 3
Estimation field
Story points
Date - 1 September 2026 - 29 September 2026
Remaining work
Number of story points left to complete this sprint
Guideline
Ideal burn rate
Story points
20
15
10
5
0
5 Sep
9 Sep
13 Sep
17 Sep
21 Sep
25 Sep
29 Sep
Date

Reflection Questions

1. Did your estimations reflect the actual effort?

Ans: Yes, the story point estimations provided a reasonable representation of the effort and complexity involved in completing the user stories. Core risk-assessment activities such as computing the credit risk tier and matching funding commitments were assigned higher points because they involve multi-factor calculations and coordination across parties. Smaller tasks such as recording loan restructuring requests or filtering marketplace listings required fewer points. The Fibonacci scale helped represent the increasing complexity and uncertainty of the different tasks.

2. Was your backlog well-prioritized?

Ans: Yes, the backlog was prioritized according to the importance of the functionality to the Micro-Lending & Peer Credit Risk Assessor platform. Loan application intake, risk scoring, and transaction/data security were given the highest priority because they are the foundational capabilities the marketplace and repayment features depend on. The peer lending marketplace, disbursement and repayment scheduling, and default management were prioritized subsequently as the platform's core operational functionality.

3. How did your simulated sprint align with your plan?

Ans: The simulated sprints aligned well with the planned backlog. Sprint 1 focused on loan application intake, credit risk scoring, and security/transaction integrity, while Sprint 2 focused on the peer lending marketplace, EMI disbursement and scheduling, and repayment monitoring and default management. The selected user stories were progressed through the Agile workflow from To Do to In Progress and finally Done. Both sprints were completed successfully, covering all planned story points.

4. What insights did the burndown chart give about your team's capacity?

Ans: The burndown chart provided a visual representation of how the remaining story points changed as work was completed during each sprint. It helped show progress against the planned workload and gave an early indication of whether the sprint's scope could realistically be completed within the one-week duration. The completed stories across both sprints also highlighted the importance of realistic story point estimation and balanced sprint planning, particularly for effort-heavy items like risk scoring and EMI schedule generation.